Ian Douglas

Senior Platform Engineer | Cloud Security & Identity

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PROFESSIONAL SUMMARY

Platform and security engineer with 15+ years building secure, code-driven infrastructure on Kubernetes, AWS, Azure, GCP, and bare metal. Ships platforms as versioned releases through gated, Git-driven pipelines. Deep experience across the full lifecycle of identity and secrets infrastructure.

TECHNICAL PROFICIENCIES

Platforms & Cloud: Kubernetes (EKS, AKS, GKE, self-managed), Docker, Custom Operators; AWS (Organizations, EC2, VPC, IAM, KMS, Secrets Manager, Lambda); Azure (AKS, VNet, Entra ID); GCP (GKE)

Identity & Security: HashiCorp Vault (PKI), OpenLDAP, Kerberos v5, SAML, Active Directory, Cloud IAM & KMS, SOPS, Policy-as-Code (OPA), NIST SP 800-207 Zero Trust

IaC & Delivery: Terraform at scale, GitOps (Flux CD), GitHub Actions, GitLab Pipelines, Jenkins

Networking: IPv6, certificate-based IPsec, BGP, VPC/VNet (Mesh, peering, VPN, Transit Gateway), DNS

Software & Data: Python, Go, Bash, PowerShell, SQL; PostgreSQL, RDS, Redis, Kafka; Prometheus, Grafana, Fluentd, Elasticsearch, Splunk; Claude Code

PROFESSIONAL EXPERIENCE

The Motley Fool | Alexandria, VA (Remote)
Senior Platform Engineer (Contract) | 04/2026 – Present

- **CI/CD Migration:** Built reusable Python tooling, developed with Claude Code agentic workflows, that migrated ~200 repositories from Jenkins to GitHub Actions, replacing a manual repo-by-repo process, and moved secrets from AWS Secrets Manager to dotenvx to decouple them from AWS.

Kings Mountain Security | Palo Alto, CA (Remote)
Co-Founder, DevSecOps Engineer & Security Architect (Part-time) | 01/2025 – Present

- **Zero Trust Architecture:** Architected and Developed Terraform-based "Zero Trust Capability Packages" modeled on NIST SP 800-207, including Security-as-Code modules to deliver DoD Zero Trust compliance in progressive phases

FlowDelta Trading | Boulder, CO (Remote)
Senior Software Engineer, Python/Data (Contract) | 01/2025 – 04/2026

- **Backtesting Platform:** Developed a proprietary backtesting platform for a boutique trading desk, including a vectorized Monte Carlo engine for multi-dimensional parameter grid sweeps.

ZagTech | San Jose, CA (Remote)
Senior Platform Engineer (Contract) | 04/2024 – 12/2024

- **Commercial Kubernetes Platform:** Engineered a proprietary Kubernetes platform on Azure (AKS), deployed across multiple enterprise clients; the firm's largest client adopted it immediately in a $250K engagement.
- **Fleet GitOps & Terraform:** Designed the Flux repository hierarchy and cluster bootstrap for AKS fleets, and a novel Terraform design for ordering complex clusters from a menu of profiles and feature sets (Kate's Kitchen).
- **Azure Networking & Cost:** Designed client connectivity (VNet peering, site-to-site VPN, hub transit routing) and rightsized infrastructure to client cost targets.

Invitae Biosciences | San Francisco, CA (Remote)
Senior Platform Engineer | 07/2023 – 04/2024

- **Research Compute Platform:** Owned operations and feature development for a genomics research platform on self-managed Kubernetes and EKS (dozens of production clusters, five environments) on a 5 person team; zero-downtime upgrades and stateful migrations, automated fleet-wide Flux bootstrap (Terraform, Go, Python).
- **Cluster Hardening:** Built a Python tool to audit every PodSecurityPolicy across clusters against the Pod Security Standards, enabling the move to Pod Security Admission (Cluster Security Audit).
- **Secrets Management Architecture:** Designed least-privilege secrets management using SOPS, AWS KMS, Shamir secret sharing.

Pearson | London, UK (Remote)
Senior Cloud Platforms Engineer | 09/2019 – 08/2021

- **Kubernetes Platform:** Operated and evolved Pearson's self-managed Kubernetes platform on AWS (own control plane, etcd, Vault PKI) for ~400 dev teams, with daily platform releases; wrote 30+ Terraform modules. The platform was load-tested to 8.5M concurrent users at 150–250 ms (case study).
- **Platform Release Pipeline:** Maintained and extended a 121-job GitHub Actions pipeline promoting 14 versioned components (full-stack Kubernetes control plane on EC2) through eight clusters to two production regions, gated by tests, automated ServiceNow change requests, and human in the loop)
- **Centralized Logging:** Built a new log aggregation service as one of two engineers (Fluentd, Python, AWS Lambda, feeding Elasticsearch and Splunk).
- **Reusable Platform & Operators:** Generalized Bitesize's code modules into a white-box platform offering, as part of a small team; co-developed Go-based operators and CRDs that provisioned most AWS services.

Stanford University | Palo Alto, CA
Windows Infrastructure Architect, then DevOps Engineer | 12/2015 – 09/2019

- **Identity Modernization:** Owned the OpenLDAP modernization within the IAM team's move of the authentication stack (Kerberos/Heimdal, Shibboleth) onto Kubernetes: master/replica StatefulSets with automated restores and lifecycle automation, running in production as the directory for the entire university.
- **Zero Trust Active Directory:** Designed and deployed Active Directory in AWS over IPv6 with certificate-authenticated IPsec, letting client VPCs with overlapping address space reach domain controllers without peering or VPN (paper).
- **Cloud Migration:** Delivered IaC and GKE-backed CI/CD that moved core services, including Puppet, to Google cloud and AWS.
- **Secrets Infrastructure:** Ran multi-tenant HashiCorp Vault as one of a three-person team: PKI automation, multi-tenant secrets store, authentication proxies.

Earlier Experience

- **Gazillion Entertainment (Marvel Heroes)**, DevOps Engineer: database and cluster-operations automation for an MMO with 40k+ daily active users and twice-weekly deployments.
- **Two Sigma Investments**, Systems Engineer: bare-metal provisioning for research and trading servers; zero-touch lifecycle for 1,000+ secure workstations.
- **Columbia University**, Systems Engineer: databases, ESX, storage, and VM infrastructure for 22 libraries.

EDUCATION

University of Denver

BA, Mathematics & Philosophy (Double Major)